Double Integrals

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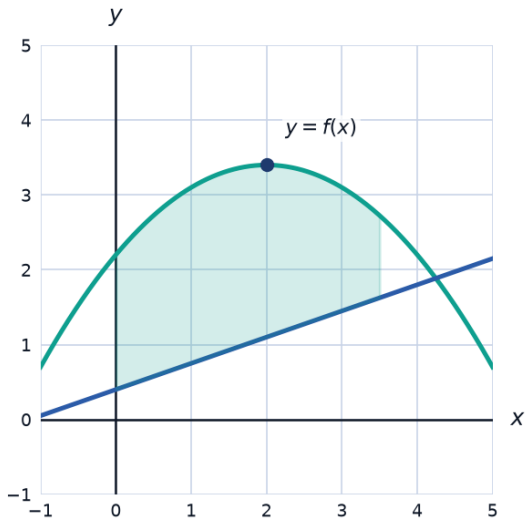
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Double integrals over a rectangle

Set up the double integral of $f(x, y) = xy$ over the rectangle $[0, 2] \times [0, 3]$ (integrate y first).

2 [Foundational] • Target: Double integrals over a rectangle

Set up the double integral of $f(x, y) = x^2 + y$ over the rectangle $[1, 3] \times [0, 2]$ (integrate x first).



3 [Foundational] • Target: Iterated integrals

Evaluate $\int_0^2 \int_0^3 xy \, dy \, dx$.

4 [Foundational] • Target: Iterated integrals

Evaluate $\int_0^1 \int_0^2 (x^2 + y) \, dy \, dx$.

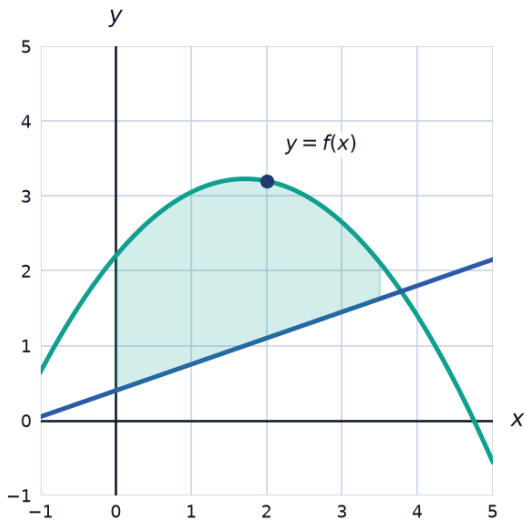
5 [Foundational] • Target: Iterated integrals

Evaluate $\int_0^\pi \int_0^1 y \sin x \, dy \, dx$.

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6 [Foundational] • Target: Fubini's Theorem: order of integration

Verify Fubini's Theorem for $f(x, y) = xy$ on $[0, 1] \times [0, 2]$ by evaluating the integral in both orders.



7 [Foundational] • Target: Fubini's Theorem: order of integration

State the hypothesis of Fubini's Theorem needed for the two iterated orders to be guaranteed equal.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Double integrals over general regions

Evaluate $\int_0^1 \int_0^x (x + y)\, dy\, dx$.

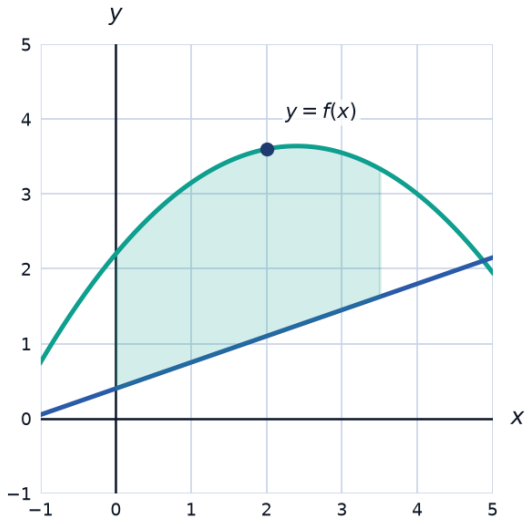
9 [Intermediate] • Target: Double integrals over general regions

Set up (do not evaluate) the double integral of $f(x, y) = xy$ over the triangular region bounded by $y = 0$, $x = 1$, and $y = x$.

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10 [Intermediate] • Target: Double integrals over general regions

Evaluate $\int_0^1 \int_y^1 2\,dx\,dy$ (a region bounded by $x = y$, $x = 1$, $y = 0$).



11 [Intermediate] • Target: Reversing the order of integration

Reverse the order of integration for $\int_0^1 \int_0^x f(x, y)\,dy\,dx$, describing the new bounds.

12 [Intermediate] • Target: Reversing the order of integration

Evaluate $\int_0^1 \int_x^1 e^{y^2}\,dy\,dx$ by first reversing the order of integration (the inner integral has no elementary antiderivative in the original order).

13 [Intermediate] • Target: Polar coordinates

Convert $\iint_D (x^2 + y^2)\,dA$ to polar coordinates, where D is the disk $x^2 + y^2 \le 4$, and evaluate.

14 [Intermediate] • Target: Polar coordinates

Evaluate $\iint_D xy\,dA$ where D is the quarter disk $x^2 + y^2 \le 1$, $x \ge 0$, $y \ge 0$, using polar coordinates.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

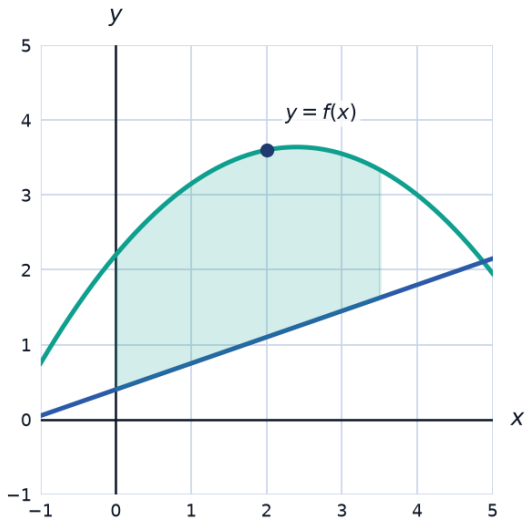
15 [Challenging] • Target: Polar coordinates • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Use polar coordinates to evaluate $\iint_D e^{-(x^2 + y^2)}\,dA$ where D is the entire plane $\mathbb{R}^2$ (the classic Gaussian integral setup).

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16 [Challenging] • Target: Area and average value • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Use a double integral to find the area of the region bounded by $y = x^2$ and $y = x$ (for $0 \le x \le 1$, where $x \ge x^2$).



17 [Challenging] • Target: Area and average value • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

The average value of $f(x, y)$ over a region D with area A is $\frac{1}{A}\iint_D f\,dA$. Find the average value of $f(x, y) = xy$ over the rectangle $[0, 2] \times [0, 3]$, using the earlier result $\iint_D xy\,dA = 9$.

18 [Challenging] • Target: Area and average value • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Use a double integral to find the area of the region bounded above by $y = 4 - x^2$ and below by $y = 0$.

19 [Challenging] • Target: Volume via double integrals • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Set up (do not evaluate) the double integral for the volume under the surface $z = x^2 + y^2$ above the rectangle $[0, 1] \times [0, 1]$.

20 [Challenging] • Target: Volume via double integrals • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Evaluate the volume integral set up above.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Fubini's Theorem over Rectangles

Evaluate $\int_0^2 \int_0^3 2xy\,dy\,dx$.

a. 18

b. 9

c. 36

d. 12

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22 [Foundational] • Target: Fubini's Theorem over Rectangles

What does Fubini's Theorem state for a continuous function $f(x, y)$ over a rectangle $R = [a, b] \times [c, d]$?

a. The iterated integrals in either order are equal: $\int_a^b \int_c^d f(x, y)\,dy\,dx = \int_c^d \int_a^b f(x, y)\,dx\,dy$

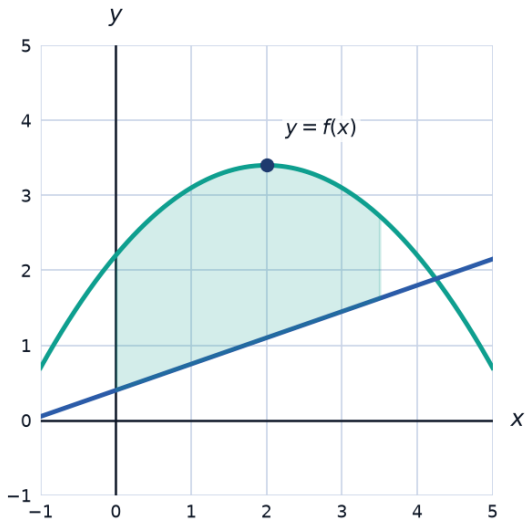
b. The double integral is always equal to $f(b, d) - f(a, c)$

c. The integral is zero if f is odd in x

d. The integral must be evaluated with x first

23 [Foundational] • Target: Fubini's Theorem over Rectangles

Evaluate $\iint_R (x + y)\,dA$ over $R = [0, 1] \times [0, 2]$.



a. 3

b. 2

c. 4

d. 1

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24 [Foundational] • Target: Double Integrals over General Regions

Set up the double integral of $f(x, y)$ over the triangular region bounded by $y = 0$, $y = 2x$, and $x = 3$ as a Type I region.

a. $\int_0^3 \int_0^{2x} f(x, y)\,dy\,dx$

b. $\int_0^6 \int_0^3 f(x, y)\,dx\,dy$

c. $\int_0^{2x} \int_0^3 f(x, y)\,dx\,dy$

d. $\int_0^3 \int_{2x}^6 f(x, y)\,dy\,dx$

25 [Foundational] • Target: Double Integrals over General Regions

Evaluate $\int_0^1 \int_0^x 6xy\,dy\,dx$.

a. $\frac{3}{4}$

b. 1

c. $\frac{1}{2}$

d. $\frac{3}{2}$

26 [Foundational] • Target: Double Integrals over General Regions

What geometric quantity does $\iint_D 1\,dA$ compute?

a. The area of the planar region D

b. The perimeter of region D

c. The volume under the paraboloid

d. The centroid

27 [Foundational] • Target: Changing the Order of Integration

To evaluate $\int_0^1 \int_y^1 e^{x^2}\,dx\,dy$, which cannot be integrated directly, reverse the order of integration.

a. $\int_0^1 \int_0^x e^{x^2}\,dy\,dx$

b. $\int_0^1 \int_x^1 e^{x^2}\,dy\,dx$

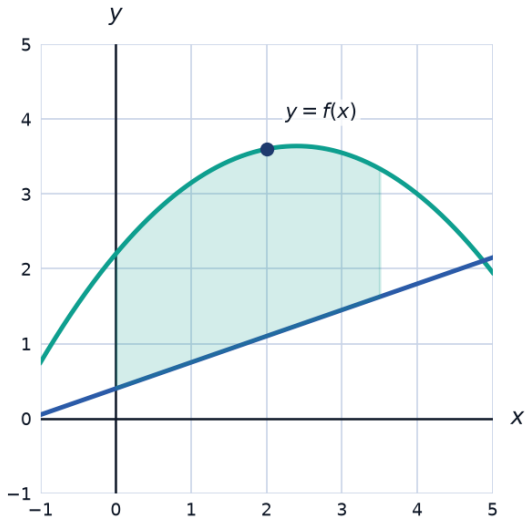
c. $\int_0^1 \int_0^1 e^{x^2}\,dx\,dy$

d. $\int_y^1 \int_0^1 e^{x^2}\,dy\,dx$

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28 [Foundational] • Target: Changing the Order of Integration

Evaluate the reversed integral $\int_0^1 \int_0^x e^{x^2} dydx$.



a. $\frac{e-1}{2}$

b. $e-1$

c. $\frac{e}{2}$

d. 1

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Changing the Order of Integration

Reverse the order of integration for $\int_0^4 \int_{\sqrt{x}}^2 f(x,y)dydx$.

a. $\int_0^2 \int_0^{y^2} f(x,y)dxdy$

b. $\int_0^2 \int_{y^2}^4 f(x,y)dxdy$

c. $\int_0^4 \int_0^{y^2} f(x,y)dxdy$

d. $\int_{\sqrt{x}}^2 \int_0^4 f(x,y)dxdy$

30 [Intermediate] • Target: Double Integrals in Polar Coordinates

What is the differential area element dA in polar coordinates?

a. $dA = rdrd\theta$

b. $dA = drd\theta$

c. $dA = r^2drd\theta$

d. $dA = \frac{1}{r}drd\theta$

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31 [Intermediate] • Target: Double Integrals in Polar Coordinates

Evaluate $\iint_D e^{-(x^2+y^2)}dA$ where D is the disk $x^2 + y^2 \leq 4$.

a. $\pi(1 - e^{-4})$

b. $2\pi(1 - e^{-4})$

c. πe^{-4}

d. $\frac{\pi}{2}(1 - e^{-4})$

32 [Intermedi... ...s in Polar Coordinates

Convert $\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2)dydx$ to polar coordinates.

a. $\int_0^{\pi/2} \int_0^1 r^3 drd\theta$

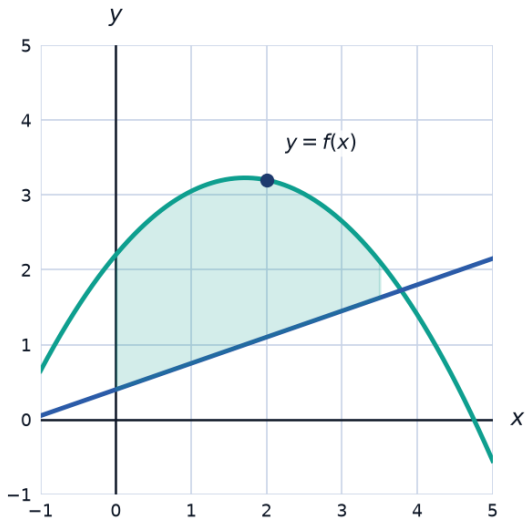
b. $\int_0^{2\pi} \int_0^1 r^2 drd\theta$

c. $\int_0^{\pi/2} \int_0^1 r^2 drd\theta$

d. $\int_0^1 \int_0^1 r^3 drd\theta$

33 [Intermediate] • Target: Double Integrals in Polar Coordinates

Evaluate the double integral of the function $f(x, y) = 2xy$ over the rectangular region $R = [0, 2] \times [0, 3]$ using Fubini's Theorem. Set up the iterated integral by integrating with respect to y first, then x.



a. 12

b. 18

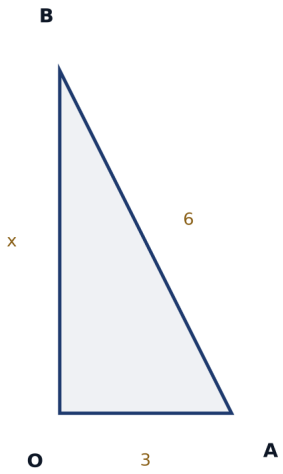
c. 6

d. 24

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34 **[Intermediate]** • **Target: Double integrals over a rectangle**

Consider the region R in the first quadrant bounded by the x-axis, the y-axis, and the line $y = -2x + 6$. We are interested in evaluating the double integral $\iint_R f(x, y)\, dA$ where $f(x, y) = x + y$. However, the integral is set up in polar coordinates as $\int_{\alpha}^{\beta} \int_{r_1(\theta)}^{r_2(\theta)} f(r\cos\theta, r\sin\theta) \cdot r\, dr\, d\theta$. If this integral is evaluated from $\theta = 0$ to $\theta = \frac{\pi}{2}$, and the inner integral bounds are $r_1(\theta) = 0$ and $r_2(\theta) = \frac{6}{\sin\theta + 2\cos\theta}$, and the outer integral is $\int_0^{\pi/2} \int_0^{\frac{6}{\sin\theta + 2\cos\theta}} (r\cos\theta + r\sin\theta) \cdot r\, dr\, d\theta$, determine the value of α and β that correspond to the description of region R in polar coordinates for $0 \le \theta \le \frac{\pi}{2}$.



a. $\alpha = 0, \beta = \frac{\pi}{2}$

b. $\alpha = 0, \beta = \frac{\pi}{4}$

c. $\alpha = 0, \beta = \frac{3\pi}{4}$

d. $\alpha = \frac{\pi}{4}, \beta = \frac{\pi}{2}$

35 **[Intermediate]** • **Target: Iterated integrals**

Consider the region R in the xy-plane defined by $0 \le x \le 1$ and $0 \le y \le x^2$. Evaluate the double integral $\iint_R y\, dA$.

a. $\frac{1}{12}$

b. $\frac{1}{10}$

c. $\frac{1}{8}$

d. $\frac{1}{6}$

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36 [Intermediate] • Target: Fubini's Theorem: order of integration

Consider the solid region E bounded below by the cone $z = \sqrt{x^2 + y^2}$ and above by the sphere $x^2 + y^2 + z^2 = 16$. The volume of E is given by the triple integral $V = \iiint_E dV$. If E is expressed in spherical coordinates, the integral becomes $\int_{\alpha}^{\beta} \int_{\gamma}^{\delta} \int_{\rho_1}^{\rho_2} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$. Determine the value of α, the lower limit of integration for θ.

a. 0

b. $\frac{\pi}{2}$

c. π

d. 2π

37 [Intermediate] • Target: Double integrals over general regions

Consider the vector field $\mathbf{F}(x, y, z) = \langle x, y, z \rangle$. Let S be the closed surface of the unit cube defined by $0 \le x \le 1, 0 \le y \le 1, 0 \le z \le 1$, oriented outwards. According to the Divergence Theorem, the flux of $\mathbf{F}$ through S is equal to the triple integral of the divergence of $\mathbf{F}$ over the volume V enclosed by S. Calculate this flux by evaluating $\iiint_V abla \cdot \mathbf{F} \, dV$.

a. 1

b. 2

c. 3

d. 4

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Reversing the order of integration • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Consider the solid region E in the first octant bounded by the coordinate planes and the surface $z = 4 - x^2 - y^2$. The volume of this region is given by the triple integral $V = \iiint_E dV$. We are given that the volume can also be expressed in cylindrical coordinates as $V = \int_0^{\pi/2} \int_0^{\sqrt{a}} \int_0^{4 - r^2} r \, dz \, dr \, d\theta$. Find the value of the parameter a.

a. 2

b. 4

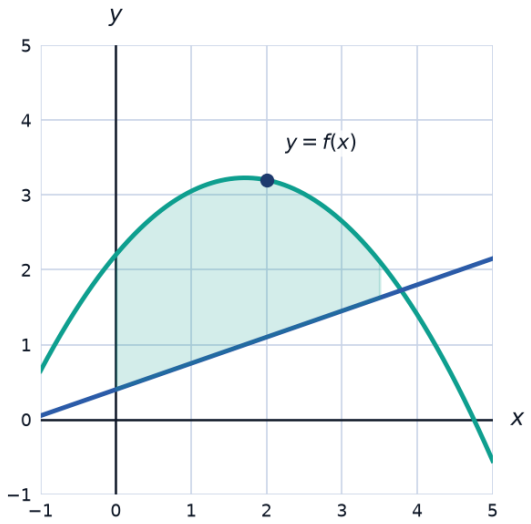
c. 16

d. 8

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39 [Challenging] • Target: Polar coordinates • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Consider the solid region E defined by $0 \le x \le 1$, $0 \le y \le 2$, and $0 \le z \le 3$. Evaluate the triple integral of the function $f(x, y, z) = 2x + 3y$ over the region E.



a. 18

b. 15

c. 21

d. 24

40 [Challenging] • Target: Area and average value • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

A solid region E is defined by the inequalities $x^2 + y^2 \le 16$, $0 \le z \le 5$. The volume of this region can be computed using a triple integral. In cylindrical coordinates, the region E is described by $0 \le r \le R$, $0 \le \theta \le 2\pi$, and $0 \le z \le H$. What is the value of R?

a. 2

b. 4

c. 8

d. 16

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41 [Challenging] • Target: Volume via double integrals • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Consider the region in the first octant bounded by the coordinate planes and the plane $x + y + z = 4$. If the density of the region is given by $\rho(x, y, z) = 3xy$, set up the triple integral for the mass of this region in Cartesian coordinates.

a. $\int_0^4 \int_0^{4-x} \int_0^{4-x-y} 3xy \, dz \, dy \, dx$

b. $\int_0^4 \int_0^{4-y} \int_0^{4-y-z} 3xy \, dx \, dz \, dy$

c. $\int_0^4 \int_0^{4-x} \int_0^{4-x-y} 3 \, dz \, dy \, dx$

d. $\int_0^4 \int_0^{4-z} \int_0^{4-z-y} 3xy \, dx \, dy \, dz$

42 [Challenging] • Target: Double integrals in applications: mass and center of mass • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Consider the vector field $\mathbf{F}(x, y) = \langle x^2, y \rangle$. According to Green's Theorem, the line integral of $\mathbf{F}$ around a simple closed curve C is equal to the double integral of $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$ over the region D enclosed by C, where $\mathbf{F} = \langle P, Q \rangle$. For the given vector field $\mathbf{F}(x, y) = \langle x^2, y \rangle$, identify the value of $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$.

a. 0

b. x^2

c. y

d. -2x

43 [Challenging] • Target: Fubini's Theorem over Rectangles • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Consider the surface S which is the portion of the plane $z = 2x + 3y + 1$ lying above the rectangular region $R = [0, 1] \times [0, 2]$ in the xy-plane. Calculate the surface area of S.

a. $\sqrt{14}$

b. $2\sqrt{14}$

c. $3\sqrt{14}$

d. $4\sqrt{14}$

44 [Challenging] • Target: Double Integrals over General Regions • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Use spherical coordinates to find the volume of the upper half of the ball $x^2 + y^2 + z^2 \leq 16$.

a. $\frac{256\pi}{3}$

b. $\frac{64\pi}{3}$

c. $\frac{128\pi}{3}$

d. 32π

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45 **[Challenging]** • **Target: Changing the Order of Integration** • **Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives**

Use spherical coordinates to find the volume of the upper half of the ball $x^2 + y^2 + z^2 \leq 4$.

a. $\frac{16\pi}{3}$

b. $\frac{32\pi}{3}$

c. $\frac{8\pi}{3}$

d. 4π

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